SAHIL THAKUR

AI / ML Engineer • Python • Machine Learning • Data Science

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PROFESSIONAL SUMMARY

Motivated AI/ML Engineer with hands-on experience building and evaluating machine learning models across regression, classification, and clustering paradigms. Skilled in the full ML pipeline — from data ingestion and preprocessing to feature engineering, model tuning, and performance evaluation. Proficient in Python, Scikit-learn, Pandas, and NumPy. Currently pursuing MCA with a focus on applying AI/ML to solve real-world problems. Eager to contribute to data-driven teams building intelligent, scalable systems.

TECHNICAL SKILLS

Languages	Python
ML / AI	Linear Regression, Decision Trees, Random Forest, KNN, SVM, K-Means Clustering, Supervised & Unsupervised Learning
Data Science	Exploratory Data Analysis (EDA), Data Cleaning, Feature Engineering, Statistical Analysis, Dimensionality Reduction
Libraries	Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn
Evaluation	RMSE, R-squared, Accuracy, Precision/Recall, Cross-Validation, Hyperparameter Tuning (GridSearchCV)
Tools	Jupyter Notebook, VS Code, Git, GitHub

WORK EXPERIENCE

Data Science Intern | **Main Flow Services** *3 Months*

- Performed end-to-end EDA on real-world datasets to uncover patterns, trends, and outliers using Matplotlib and Seaborn.
- Cleaned and transformed raw datasets using Pandas and NumPy, improving data quality and consistency for downstream modeling.
- Built and benchmarked supervised ML models (Linear Regression, Decision Tree) using Scikit-learn with measurable accuracy improvements.
- Applied k-fold cross-validation and hyperparameter tuning to reduce overfitting and improve model generalization.
- Documented findings and model results in structured Jupyter Notebooks for team review and reproducibility.

Python Full Stack Developer Intern | **Key Resolvent Pvt Ltd** *4 Months*

- Developed and optimized Python-based backend modules, improving application reliability and response performance.
- Integrated relational databases with backend services; optimized SQL queries for data-intensive features.
- Participated in debugging, testing, and deployment cycles for full-stack web applications in an agile environment.
- Gained practical experience with structured data handling and backend-to-frontend data pipelines.

PROJECTS

House Price Prediction Model — *Machine Learning · Scikit-learn · Python*

- Designed and trained a supervised regression pipeline using Linear Regression and Decision Tree on structured housing datasets.

- Conducted feature selection and EDA; handled missing values, encoded categorical variables, and normalized features.
- Evaluated models with RMSE and R-squared metrics; comparative analysis identified the best-performing algorithm.
- Tuned hyperparameters using GridSearchCV to optimize prediction accuracy and minimize overfitting.

Vehicle Management System — *Python Application · Data Management · OOP*

- Built a Python-based parking management system tracking vehicle entry and exit timestamps with accurate record keeping.
- Implemented structured data storage and retrieval logic for efficient, reliable lookups and reporting.
- Designed a CLI interface using OOP principles for maintainable, extensible, and user-friendly operation.

EDUCATION

Master of Computer Applications (MCA) — Glocal University *2024 – 2026 (Pursuing)*

Bachelor of Computer Applications (BCA) — GGSIPU University *2021 – 2024*